

Online controller selection for the angular control of gravitational-wave detectors under non-stationary noise: a multi-armed bandit approach

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Draft — July 10, 2026

Abstract

The angular control loops of gravitational-wave (GW) detectors must reject radiation-pressure (Sidles–Sigg) torques whose strength, together with the seismic and sensing-noise environment, varies over time. In current operation the choice of control filter is changed by hand by operators, for instance switching from low- to high-gain alignment control when an earthquake early-warning system forecasts elevated ground motion. We formulate this choice as an *online controller-selection* problem and study whether a multi-armed bandit policy can perform the selection autonomously so as to minimise a long-horizon cost. Using a time-domain simulator of the pitch degree of freedom of a coupled-cavity detector, validated against a reference implementation, we define three operating regimes and a family of fixed linear controllers with a designed one-to-one regime–controller correspondence, and a scalar reward built from the hard-eigenmode error and control effort. We make three contributions ahead of the bandit study itself. First, we characterise the reward landscape and show that the designed one-to-one regime–controller mapping holds in all three regimes, robustly in the nominal and intermediate regimes and marginally in the extreme regime, with a non-saturating reward. Second, we quantify the cost of switching controllers and find a pronounced asymmetry: gain-decreasing switches are far costlier than gain-increasing ones. Third, and most importantly, we show that the apparent switching penalty is dominated by the *controller-handover implementation*: a naive state reset or raw state carry-over produces large transients, whereas a bumpless, parallel-bank handover renders the switching cost negligible for this gain-scaled controller family. Finally, we introduce *Recurrence-Aware Thompson Sampling* (RA-TS-F), a minimal policy combining Thompson sampling with a drift detector, a library of per-regime posteriors that are *resumed* when a regime recurs, and a bounded coverage probe. On one-week and six-month simulated benchmarks RA-TS-F attains the lowest regret of all policies tested—non-stationary bandit baselines, an operator-like rule-based policy, and every fixed controller—capturing 74–83% of the oracle’s achievable advantage over the best fixed controller, with a regret $4\text{--}7\times$ lower than every alternative over six months.

1 Introduction

Second-generation gravitational-wave detectors such as Advanced LIGO [1] and Advanced Virgo [2] are kilometre-scale Fabry–Perot–Michelson interferometers whose test masses are suspended as multi-stage pendula. Beyond the longitudinal degrees of freedom used to read out the GW signal, the *angular* degrees of freedom of the mirrors must be actively controlled:

residual angular motion couples to the longitudinal signal and, more fundamentally, the circulating optical power exerts radiation-pressure torques that soften one angular eigenmode and stiffen the other—the Sidles–Sigg effect [3]. The alignment sensing and control (ASC) loops that stabilise these modes operate close to a power-dependent instability and are a recognised limit to detector duty cycle and low-frequency sensitivity [4].

The disturbance environment of these loops is strongly non-stationary. Ground motion exhibits a diurnal cycle, a microseismic band that is modulated by weather and oceanic activity on timescales of hours to days, and rare, intense transients associated with earthquakes and storms. A single fixed controller cannot be simultaneously optimal across this range of conditions: a high-gain loop that rejects large low-frequency disturbances injects excess control noise when the environment is quiet, while a low-gain loop that is quiet in nominal conditions fails to maintain alignment during elevated ground motion. In current practice this trade-off is managed manually: operators (or simple rule-based logic) switch between predefined control filters, for example moving to a high-gain alignment configuration when an earthquake early-warning system predicts high incoming ground motion.

This manual, forecast-driven switching is a natural candidate for automation. We pose it as an *online controller-selection* problem: given a fixed library of controllers and a scalar performance signal computed online from the loop’s own error and control signals, an autonomous policy must repeatedly choose which controller to apply so as to minimise the time-integrated cost, without prior knowledge of when the environment will change. This is precisely the setting of the *non-stationary multi-armed bandit* [5, 6, 7]: each controller is an arm, the negative cost is a stochastic reward whose distribution shifts when the environment changes, and the policy must balance exploration (testing controllers to detect such shifts) against exploitation (using the controller currently believed best).

In this work we build the simulation and problem infrastructure required to study this question rigorously, and we report a set of results that are prerequisites to—and partly independent of—the choice of bandit algorithm. Our contributions are:

1. A time-domain detector model with an explicit, validated plant, in which three operating regimes and a family of fixed linear controllers are defined so as to admit a designed one-to-one regime–controller mapping (Sections 3–4).
2. A characterisation of the resulting reward landscape (Section 5), establishing that the one-to-one mapping holds in all three regimes (robustly in the nominal and intermediate regimes, marginally in the extreme regime) and that the reward is non-saturating, so a learnable signal is present in every regime.
3. A quantitative study of the *dynamics of switching* (Section 6): the transient cost of changing controllers, its asymmetry in the direction of the gain change, and—central to the feasibility of online switching—the dependence of this cost on the controller-handover implementation, with the conclusion that a realistic bumpless handover makes switching nearly free.
4. An online-selection policy, *Recurrence-Aware Thompson Sampling with a coverage floor* (RA-TS-F), designed around the structure established in 1–3: regimes recur, their identity is carried by the *shape* of the per-arm rewards rather than their level, and switching is cheap. On one-week and six-month benchmarks with a common noise realisation across policies, RA-TS-F attains the lowest regret of every policy tested (Section 7).

2 Background

2.1 Angular control and the Sidles–Sigg effect

Consider the two test masses forming a single arm cavity of length L with radii of curvature R_1 (input, ITM) and R_2 (end, ETM). In the pitch (vertical-plane tilt) degree of freedom, the cavity geometry is described by the g -factors $g_i = 1 - L/R_i$. The radiation pressure of the circulating power couples the two mirror tilts; diagonalising the coupling yields two angular eigenmodes, conventionally termed *soft* and *hard*, with optical-spring-modified torsional stiffnesses. The transformation from the local (ITM, ETM) tilt basis to the eigenbasis is

$$r = \frac{1}{2} \left(g_1 - g_2 + \sqrt{(g_1 - g_2)^2 + 4} \right), \quad T_{\text{loc} \rightarrow \text{eig}} = \frac{1}{1 + r^2} \begin{pmatrix} 1 & r \\ -r & 1 \end{pmatrix}, \quad (1)$$

and the beam-spot-to-angle response factors of the two modes are

$$\left. \frac{\partial y}{\partial \theta} \right|_{\text{soft/hard}} = \frac{L}{2} \frac{(g_1 + g_2) \pm \sqrt{(g_2 - g_1)^2 + 4}}{g_1 g_2 - 1}. \quad (2)$$

For the parameters of interest the hard mode is the limiting loop: it carries the dominant radiation-pressure torque and is the relevant control-noise injection path, and we therefore use it as the sole performance channel throughout (Section 4). The circulating power that sets the strength of this coupling follows the Fabry–Perot resonance condition

$$P_{\text{cav}} = \frac{P_{\text{in}} T_1}{\left| 1 - \sqrt{R_1 R_2} \exp(4\pi i \delta L_{\text{hp}} / \lambda) \right|^2}, \quad (3)$$

with T_1 the ITM power transmissivity, λ the laser wavelength, and δL_{hp} the high-passed cavity-length change induced by the mirror tilts.

2.2 The non-stationary disturbance environment

The seismic input to the suspensions varies on multiple timescales: a diurnal component, a micro-seismic band modulated by weather over hours to days, and sporadic high-amplitude transients. Operationally, this motivates a small set of qualitatively distinct *regimes* and a corresponding set of controllers tuned for each, between which the configuration is switched. A key methodological point, which we adopt throughout, is that physically relevant non-stationarity is *adiabatic*: the environment evolves continuously rather than by abrupt steps, so that the interesting question is whether control remains stable, and a selection policy beneficial, *through* the slow transitions [11]. **[TODO: cite the LIGO/MIT operational practice of earthquake-triggered low/high-gain ASC switching with a proper reference.]**

2.3 Bandits for non-stationary rewards

In the stochastic multi-armed bandit, a policy repeatedly selects one of K arms and observes a reward drawn from that arm’s (unknown) distribution, with the goal of minimising regret relative to the best arm [5]. When the reward distributions change over time the relevant notion is *non-stationary* regret against the best arm *at each time*. Two broad families address this: passively forgetting methods, such as discounted or sliding-window UCB [6] and exponential-weighting variants; and actively adaptive methods that detect distribution changes, of which AdSwitch [7] is a representative with regret guarantees under an unknown number of changes. Under milder, more informative statistical assumptions about the regimes, Bayesian approaches such as Thompson sampling [8, 9], possibly in a contextual form, may exploit structure that the worst-case methods cannot. The appropriate choice depends on the regime-switching statistics

and the separation of the per-arm reward distributions; quantifying the latter is one purpose of Section 5. A third structural feature, central to our setting, is *recurrence*: the environment does not drift to ever-new states but revisits a small set of regimes. A policy that re-learns each regime from scratch on every visit pays the identification cost repeatedly; a policy that recognises a returning regime and resumes previously accumulated knowledge pays it once. Exploiting recurrence, with change detection in the spirit of CUSUM-type tests [10], is the design principle behind the policy of Section 7.3.

3 Simulation framework

3.1 Time-domain plant model

We use a time-domain simulator (“Lightsaber”) of the pitch degree of freedom of a single arm cavity, run as a per-sample closed loop at a sampling frequency $f_s = 256$ Hz. The principal parameters are an arm length $L = 3994.5$ m, radii of curvature $R_1 = 1934$ m and $R_2 = 2245$ m, ITM transmissivity $T_1 = 0.014$, laser wavelength $\lambda = 1064$ nm, and a nominal arm-cavity input power $P_{\text{in}} = 705$ W (corresponding to a nominal circulating power of order 200 kW).

At each time step the plant (i) injects test-mass pitch noise obtained by filtering colored seismic and damping-loop (OSEM) noise through measured suspension transfer functions; (ii) computes the cavity power from Eq. (3); (iii) forms the radiation-pressure torques on the two masses,

$$\tau = \frac{2}{c} \left(P_{\text{cav}} \mathbf{s} + \mathbf{s}_0 (P_{\text{cav}} - P_{\text{dc}}) \right), \quad (4)$$

with $\mathbf{s}$ the beam-spot vector, $\mathbf{s}_0$ a static beam-spot offset defining the operating point, and P_{dc} a reference power; (iv) propagates torque and actuation to mirror angle through the radiation-pressure (TM→P) and actuator (PUM→TM) state-space models; (v) reads out the two eigenmodes via Eq. (1) with additive sensing noise; (vi) applies a power-dependent Sidles–Sigg compensation path; and (vii) computes the controller actuation. The controllers act on the eigenmode error and their output is mapped back to the local actuation basis. All recursive elements (plant, compensator, controller) are second-order-section (SOS) digital filters with persistent internal state, so that the simulation is a faithful continuous closed loop rather than a sequence of independent batches.

3.2 Validation against the reference simulator

The plant used here is a refactored version of an earlier, component-graph reference implementation. We verified that the refactoring preserves the physics. The two core plant transfer functions—the actuator response (PUM→TM, gain 93.53) and the radiation-pressure response (TM→P, gain 2.5677)—are identical zero–pole–gain models; the three measured suspension transfer functions are bit-identical between the two codebases (maximum amplitude difference 0); and the high-pass, Fabry–Perot, eigenmode and beam-spot expressions coincide. The differences between the two implementations are confined to the deliberately modified experimental layer—the static operating point $\mathbf{s}_0$, the form of the Sidles–Sigg compensator, the controller family (Section 4), and the regime-dependent noise inputs—and none constitutes a change to the underlying plant. We therefore treat the present plant as physically equivalent to the reference. **[TODO: Optionally add a quantitative dynamic cross-check (matched-configuration output PSD overlay).]**

Table 1: Regime definitions: multiplicative scale factors applied to the nominal configuration.

	R_0 (nominal)	R_1 (high-micro)	R_2 (extreme)
Optical-power scale	1.00	0.95	0.90
Sensing-noise scale	120	12	0.25
Seismic-input scale	1.0	6.0	20.0
OSEM (damping) scale	1.0	1.0	1.0

4 Problem formulation

4.1 Operating regimes

We define three operating regimes that differ along physically motivated axes: the optical power (and hence the Sidles–Sigg coupling strength), the seismic input level, and the sensing (readout) noise. The scale factors, applied to the nominal configuration, are listed in Table 1; the underlying amplitude spectral densities are derived from O4-era data. Regime R_0 is the nominal, low-ground-motion state; R_1 a high-microseism state; and R_2 an extreme state with strongly elevated seismic input but a comparatively clean sensor.

4.2 Controller family

We consider three fixed hard-mode controllers, C_0 , C_1 , C_2 , which share an identical filter shape and differ only by an overall DC gain (30, 44.2, 50 respectively). They are therefore a one-parameter gain family: driven by the same error signal, their outputs are proportional, $u^{(j)} = (g_j/g_i)u^{(i)}$. The goal is to *select* among them online, not to retune them. The soft-mode controller is common to all three and is not varied.

4.3 Observables and scalar reward

Performance is assessed on the hard eigenmode only, through two observables: the hard-mode error signal e_h (the controller input/readout) and the hard-mode control effort u_h (the controller output, transformed to the eigenbasis). A linear scalarisation of error and control, $J = \alpha \text{RMS}(e_h) + \beta \text{RMS}(u_h)$, cannot yield a consistent regime–controller preference, because different regimes call for opposite error/control trade-offs; a fixed trade-off slope therefore fails. We instead use a multiplicative (logarithmic) cost. In its minimal form,

$$J = \rho \log \frac{\text{RMS}(e_h)}{\min_C \text{RMS}(e_h)} + \log \frac{\text{RMS}_{10-30\text{Hz}}(u_h)}{\min_C \text{RMS}_{10-30\text{Hz}}(u_h)}, \quad (5)$$

with a single global coefficient $\rho \approx 1.095$, which already yields the intended one-to-one mapping. In the implementation used for the bandit reward we adopt a generalised, multi-band form: the error and control signals are decomposed into a set of Butterworth frequency bands, and the reward is a weighted sum of the log band-powers evaluated on 2 s blocks,

$$Z = \sum_i w_i \log b_i, \quad r_{\text{raw}} = -Z, \quad (6)$$

with band features b_i (RMS per band of e_h and u_h) and fixed weights w_i calibrated so as to reproduce the designed mapping. Because the downstream policy expects a bounded reward, the per-segment mean raw score is mapped to $[0, 1]$ by a logistic,

$$r = [1 + \exp(-(\bar{r}_{\text{raw}} - r_0)/\sigma_0)]^{-1}, \quad r_0 = 167.5, \sigma_0 = 2.5. \quad (7)$$

The implications of this normalisation are examined in Section 5.

Table 2: Mean reward per (regime, controller). Raw score r_{raw} [Eq. (6)] and logistic (bandit-space) reward [Eq. (7)]. Bold marks the per-regime optimum.

	raw r_{raw}			logistic r		
	C_0	C_1	C_2	C_0	C_1	C_2
R_0	168.59	167.75	167.38	0.597	0.521	0.487
R_1	165.52	165.68	165.39	0.325	0.338	0.314
R_2	167.43	168.23	168.28	0.492	0.565	0.569

4.4 Designed regime–controller mapping

The regimes and controllers are constructed so that, under the cost of Eq. (5)–(6), each regime has a distinct optimal controller: $R_0 \rightarrow C_0$, $R_1 \rightarrow C_1$, $R_2 \rightarrow C_2$. Physically, the noisy sensor of R_0 favours the low-gain controller (which injects less control noise), while the clean sensor and elevated seismic input of R_2 favour the high-gain controller (which rejects the larger disturbance). The accuracy of this mapping is the subject of Section 5.

5 Characterisation of the reward landscape

We characterise the reward of every controller in every regime by simulating all nine (regime, controller) combinations. The band features are obtained by filtering each error/control signal *continuously* (the band-pass filter state is carried across the run; the lowest band, 0.05–0.3 Hz, carries the largest weight and requires this), and the reward of Eq. (6) is the mean over 2 s blocks. Statistics are accumulated over five independent noise realisations (seeds) of 360 s each; discarding a 60 s filter warm-up per realisation leaves 149 settled 2 s blocks per seed, 745 pooled per cell in total. This pooled set of 745 per-cell blocks is what the histograms of Fig. 1 (top) show. Because the 149 blocks within one realisation come from a continuously-stateful filter, consecutive blocks are autocorrelated and cannot be treated as independent draws for an error estimate; we therefore first average the 149 blocks of each realisation down to a single per-seed mean reward, and report the bar heights and error bars of Fig. 1 (bottom) as the mean and standard error of *those five per-seed means* (five genuinely independent samples), not of the 745 raw blocks. Table 2 reports the corresponding mean raw and logistic-mapped reward.

Three features stand out. First, the designed one-to-one mapping holds in *all three* regimes: $R_0 \rightarrow C_0$, $R_1 \rightarrow C_1$ and $R_2 \rightarrow C_2$ are each the per-regime optimum (Table 2). Second, the separation is regime-dependent. It is large and highly robust in the nominal regime R_0 (raw gap 0.84, $\sim 35\sigma$ over noise realisations) and robust in the intermediate regime R_1 (gap 0.16, $\sim 4.5\sigma$), but marginal in the extreme regime R_2 , where C_1 and C_2 are nearly equivalent (gap 0.05, $\sim 0.8\sigma$). Because C_1 and C_2 are close in R_2 , the per-step regret of confusing them is small and contributes little to the integrated cost. Third, with the corrected continuous-filter reward the logistic normalisation of Eq. (7), centred at $r_0 = 167.5$, is *non-saturating* over the reward range (≈ 165 – 169): the bandit-space rewards span 0.31–0.60 across all regimes (Fig. 1, bottom), so the per-regime separation is preserved rather than compressed. A learnable signal is therefore present in every regime—strongest in the nominal regime, weakest in the extreme regime—which is what makes the online-selection problem well posed.

Fig. 1 (top) adds the distributional view that matters for the online policy: the per-block reward distributions of the three controllers overlap almost entirely in every regime. In the bandit-reward space the per-window ($T_{\text{hold}} = 100$ s) standard deviation is $\hat{\sigma} \approx 0.041$, comparable to the R_1 gap (0.013) and much larger than the R_2 gap (0.005): resolving the best arm in the hardest regime requires averaging tens of windows per arm. Equally important, the achievable reward *levels* of R_0 and R_2 are similar (Table 2); what distinguishes the regimes is the *order-*

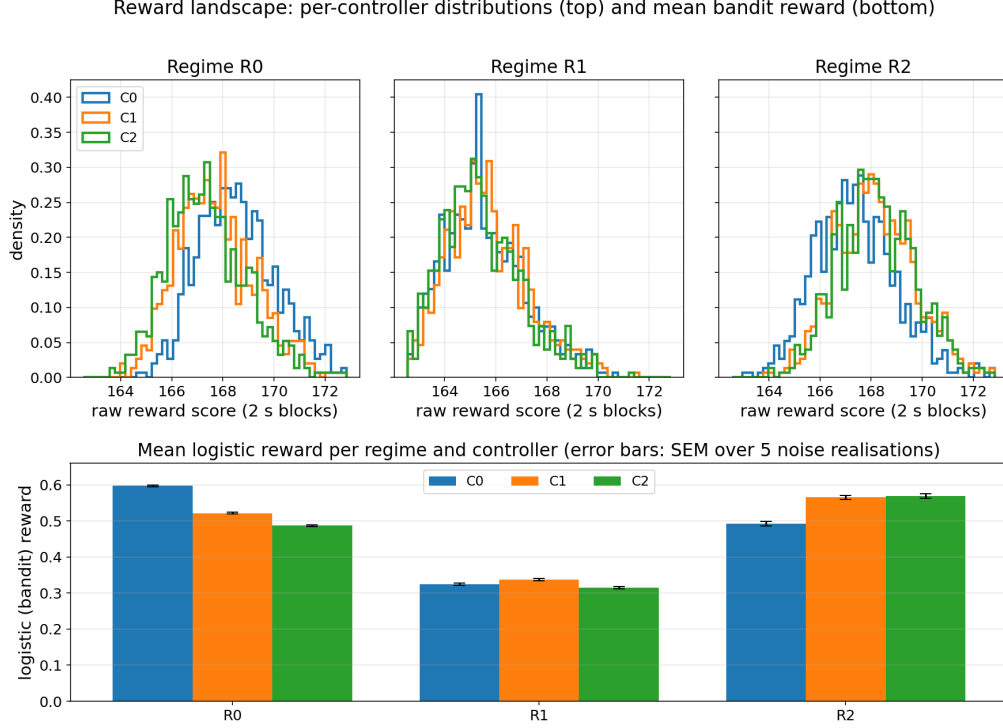


Figure 1: Reward landscape: the same data at two levels of averaging. *Top*: distribution of *individual* 2s-block raw scores per controller and regime (745 blocks per cell = 5 independent 360s noise realisations $\times$ 149 settled blocks each). *Bottom*: each realisation is reduced to its mean reward (in the logistic bandit space); bars show the average of the five realisation means, error bars their standard error. The contrast between the broad histograms and the small error bars is the averaging: a single block is noise-dominated ($\sigma \approx 1.6$, larger than every regime gap), while a 149-block run mean is 10–30 $\times$ tighter. The ordering—invisible block to block—is thus resolved decisively in R_0 , clearly in R_1 , and only marginally in R_2 , where C_1 and C_2 remain statistically indistinguishable.

ing and spacing of the three controllers—the shape of the per-arm mean-reward vector. Both observations directly shape the algorithm design of Section 7.

6 Dynamics of controller switching

A controller switch perturbs the loop transiently. To quantify this cost in isolation from the (separate) steady-state reward difference between controllers, we use a common-random-numbers design. For a transition $C_i \rightarrow C_j$ we run a *switch* simulation (C_i for a warm-up T_w , then C_j) and an *always- C_j* reference on the *identical* noise realisation, and form the post-switch reward difference

$$\Delta r(t) = r_{\text{switch}}(t) - r_{\text{always-}C_j}(t). \quad (8)$$

After the switch both runs apply C_j on the same noise, so the steady-state reward of C_j cancels and Δr isolates the transient; that $\Delta r(t) \rightarrow 0$ at long times (Fig. 2) confirms the cancellation. We report the dip depth $\min_t \Delta r$ and the integrated loss $\int \Delta r dt$, averaged over six seeds, with $T_w = 64$ s (long enough for the continuous reward filters to settle) and a 40s test window. All runs use the validated fast kernel and the corrected continuous-filter reward.

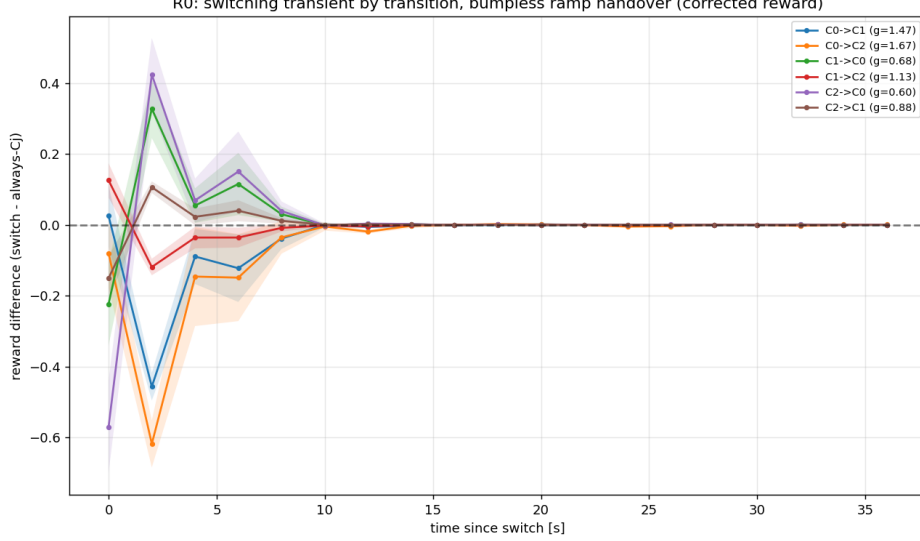


Figure 2: Switching transient $\Delta r(t)$ in R_0 for all six transitions, with the realistic bumpless-ramp handover. The dip is shallow ($\lesssim 0.6$ in reward) and recovers within a few seconds for every transition, so the switch cost is small under a proper handover.

6.1 Transient cost and its asymmetry

A switch produces a brief reward dip; the loop remains stable throughout (no divergence). With the *naive* handovers the cost is strongly asymmetric in the direction of the gain change: gain-*decreasing* switches (from a higher- to a lower-gain controller) are far costlier than gain-increasing ones, the extreme case in R_0 being $C_2 \rightarrow C_0$ (the largest gain drop), which reflects the abrupt re-equilibration of the loop when the effective gain changes discontinuously. With the realistic handovers this asymmetry is largely removed (Fig. 2): every transition incurs only a shallow, few-second dip.

6.2 Dependence on the handover implementation

The magnitude of the transient depends far more on *how* the controller is handed over than on the transition itself. We compare four implementations of the switch:

- **naive carry-over (“hot”)**: the outgoing controller’s internal filter state is transferred verbatim into the incoming controller. Because the gain difference resides in an internal filter section, the carried state is mis-scaled and the transient is *worsened*.
- **state reset (“cold”)**: the incoming controller’s filter state is zeroed. In a digital control system this produces an output discontinuity and an actuator impulse, and is not representative of switching a loop that must remain locked.
- **parallel banks**: all candidate controllers run continuously on the same error, each maintaining its own state; only the active output drives the suspensions. A switch changes which output is applied, producing a clean (g_j/g_i) step in actuation.
- **bumpless ramp**: as parallel banks, but the applied command is cross-faded between the outgoing and incoming controller outputs over a short window, $u(t) = (1 - \alpha)u^{(i)}(t) + \alpha u^{(j)}(t)$ with $\alpha : 0 \rightarrow 1$. For this gain family the blend is exactly a continuous slew of the effective loop gain from g_i to g_j , and is stable provided the loop has gain margin across the range. This models the gain-ramped, parallel-filter-bank handover used in detector digital-control systems.

Table 3: Switching cost for $C_2 \rightarrow C_0$ in R_0 (gain ratio 0.60) by handover implementation. Integrated loss in (raw score)·s; dip in raw score.

Handover	integrated loss	dip depth	realism
naive carry-over (hot)	−140	−20.3	implementation artifact
state reset (cold)	−8.4	−3.3	actuator impulse; unrealistic in lock
parallel banks	−5.1	−2.2	realistic; gain-step
bumpless ramp, 8 s	−2.8	−1.1	realistic; smooth
bumpless ramp, 2 s	+0.2	−0.6	realistic; near-eliminated

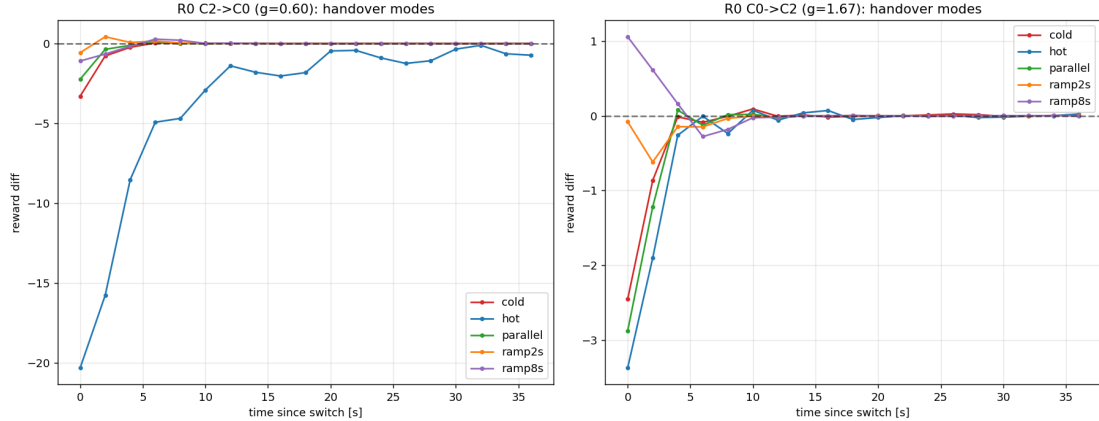


Figure 3: Switching transient by handover implementation, for $C_2 \rightarrow C_0$ (left) and $C_0 \rightarrow C_2$ (right) in R_0 . A bumpless ramp (orange/purple) suppresses the transient that the naive carry-over (blue) and, to a lesser extent, the state reset (red) and parallel-bank (green) handovers exhibit.

Table 3 and Fig. 3 compare these on the worst relevant case, $C_2 \rightarrow C_0$ (a switch into the R_0 optimum). The integrated loss spans more than two orders of magnitude: the naive carry-over is by far the worst, state reset and parallel banks are comparable and moderate, and a short (~ 2 s) bumpless ramp very nearly eliminates the transient.

The central conclusion is that the apparent switching penalty is, for this controller family, largely an artifact of the handover *implementation* rather than an intrinsic physical cost. With a bumpless, parallel-bank handover the per-switch transient (integrated loss $\lesssim 1$ in reward·s; a few-second dip of $\lesssim 0.6$) is small: averaged over a hold window of hundreds of seconds it shifts the window-mean reward by far less than the per-regime separation between controllers (Section 5), so an online policy can switch freely. Two qualifications apply. First, this relies on the controllers forming a gain family, for which the cross-fade is provably a smooth gain slew; for controllers of different shape—e.g. future learned controllers—a convex blend of two stabilising controllers is not guaranteed to be stabilising mid-ramp, and the switching cost must be re-measured. Second, even an ideal handover cannot remove the physical re-equilibration of the loop to a different closed-loop bandwidth; the recovery times of a few seconds reflect this irreducible component.

7 Online controller selection

7.1 The decision problem and its difficulties

The policy observes, once per decision window of $T_{\text{hold}} = 100$ s, a single number: the window-mean reward of the controller it chose to play. It observes *nothing else*—no regime label, no seismometer channel, no forecast. (Only the reference Oracle is given the true regime mixture.)

On this information it must repeatedly choose the next controller. The characterisation of Sections 5–6 makes the problem concrete, and hard, in five specific ways.

1. **Partial observability of the regime.** The regime identity is carried by the *shape* of the per-arm mean-reward vector (Fig. 1), but a policy playing one arm per window sees one coordinate of that vector at a time. Worse, the achievable reward *levels* of R_0 and R_2 nearly coincide, so a change $R_0 \leftrightarrow R_2$ can be invisible in the reward stream of the currently played arm: detecting it may require playing arms that are currently believed suboptimal.
2. **Noise-to-gap ratio.** The per-window reward noise ($\hat{\sigma} \approx 0.041$) exceeds the R_1 arm gap (0.013) and dwarfs the R_2 gap (0.005). Identification therefore costs many windows, and an algorithm that re-identifies from scratch at every regime visit pays this cost at every visit.
3. **Recurrence.** The environment revisits the same three regimes (Section 2.2) hundreds of times over months. This is the structure a good policy must exploit: knowledge accumulated in past visits of a regime remains valid and should be *resumed*, not re-learned.
4. **Continuous transitions.** Regime changes are ramps, not jumps (the adiabatic principle of Section 2.2). During deep transitions the reward of *every* arm collapses transiently; a change detector tuned only for level drops can mistake the collapsed, uniformly flat reward landscape for a new stationary regime and corrupt its statistics—a failure mode we repeatedly observed in earlier design iterations of the policy.
5. **Short episodes.** R_1 storms last tens of minutes: an identification procedure whose budget exceeds the episode length never learns that regime at all, no matter how statistically sound it is.

Two facts work in the policy’s favour. Switching is nearly free under the bumpless handover (Section 6), so exploration is cheap at the actuation level; and the number of distinct regimes is small, so a memory of per-regime statistics is compact. The design question is therefore not “how to forget the past” (the premise of discounted methods) but “how to *file* the past under the right regime and retrieve it on recurrence.”

7.2 Policies compared

We benchmark three groups of policies, all consuming the identical reward stream defined above.

References. Each **fixed controller** C_0, C_1, C_2 ; an **Oracle** that is given the true regime mixture at each decision and plays the argmax of the calibrated reward table (upper bound); and a **rule-based** policy modelling current operator practice: an exponentially weighted moving average of the observed reward triggers an alarm when it stays below a calibrated floor, a routine re-check fires periodically regardless, and either trigger starts a short round-robin probe of all controllers, committing to the best measured. By construction this baseline is blind to level-preserving regime changes (difficulty 1) except through its periodic re-check.

Passively forgetting bandits. Discounted UCB [6] and discounted (Gaussian) Thompson sampling [8, 9], both with discount $\gamma = 0.999$; a sweep over γ did not qualitatively change their behaviour. These treat non-stationarity as generic drift and carry no regime memory.

The proposed policy. Recurrence-Aware Thompson Sampling (Section 7.3), together with three ablations that isolate its design choices.

7.3 Recurrence-Aware Thompson Sampling (RA-TS-F)

RA-TS-F is a deliberately minimal answer to the difficulty list of Section 7.1: *just* Thompson sampling, a drift detector, and a library of per-regime posteriors, arranged so that regime identification *emerges* from Thompson sampling’s own exploration—with no scheduled probing, no separate identification phase, and no statistical certification step. (We arrived at it by successively removing exactly such machinery from earlier design iterations, each removal improving both simplicity and regret.) It has two internal modes.

TRACK. The policy believes the environment is in a known regime g and runs standard Gaussian Thompson sampling on that regime’s *accumulated* posterior: for each arm a , with $n_{g,a}$ past rewards summing to $s_{g,a}$, the posterior over the mean reward is the conjugate

$$\mu_{g,a} | \mathcal{D} \sim \mathcal{N}\left(\frac{\mu_0/\tau_0^2 + s_{g,a}/\hat{\sigma}^2}{1/\tau_0^2 + n_{g,a}/\hat{\sigma}^2}, \left(1/\tau_0^2 + n_{g,a}/\hat{\sigma}^2\right)^{-1}\right), \quad (9)$$

and the played arm is the argmax of one sample per arm. Exploitation therefore becomes near-deterministic as evidence accumulates—*without* any explicit commit step—while a persistent small exploration probability remains wherever arms are genuinely close (as in R_2), which is exactly where exploration stays cheap. Each observed reward also feeds a per-arm CUSUM [10] drift statistic

$$G_a \leftarrow \max(0, G_a + |r - b_a| - \nu), \quad \text{alarm when } G_a \geq h, \quad (10)$$

where b is a *frozen snapshot* of the regime’s per-arm means taken when the regime was adopted (so the detector cannot blind itself by tracking the drift it is meant to detect), $\nu = 1.3\hat{\sigma}$ is a slack absorbing noise, and $h = 8\hat{\sigma}$ sets the detection-delay/false-alarm trade-off. An alarm empties the working state and enters ACQUIRE; the library entry keeps its evidence.

ACQUIRE. After an alarm (or at start-up) the working posterior is reset to a diffuse prior ($\tau_0 = 0.5$), so Thompson sampling naturally rotates across arms. A *bounded coverage floor*—the “-F” in RA-TS-F—plays the least-sampled arm whenever some arm has fewer than $n_{\min} = 3$ samples; this is the entire forced-exploration budget, at most $n_{\min}K = 9$ windows per re-acquisition. Once covered, the centred shape of the acquisition estimate, $\psi = \hat{\mu} - \bar{\mu}$, is compared with the centred shape of every library regime; centring cancels any common reward level, making recognition level-invariant by construction (difficulty 1). The nearest regime is *resumed*—its posterior restored and the acquisition samples folded in—only if it matches within a tolerance ($0.8\Delta_{\text{sh}}$, with $\Delta_{\text{sh}} \approx 0.053$ the calibrated minimum shape separation between regimes) *and* beats the runner-up by a margin ($0.4\Delta_{\text{sh}}$); a shape separated from every known regime, once matured ($n_{\text{new}} = 8$ samples per arm), founds a new library entry. There is no scheduled probe, no certification test, no explicit level estimate, and no assumed number of regimes.

Why this works here can be read off the difficulty list: recurrence is exploited by resuming posteriors, so the noise-to-gap cost (difficulty 2) is paid once per regime rather than once per visit; level-invariant shape matching handles the level-degenerate R_0/R_2 pair; the coverage floor guarantees the few loser-arm samples that identification needs; and because acquisition is Thompson sampling itself, the exploration performed while identifying is simultaneously near-optimal exploitation—identification is not a separate tax. The floor is not optional: plain RA-TS (identical but with the floor disabled) *stalls*, because Thompson sampling’s very purpose is to stop playing losing arms, so arm coverage is never reached, identification never completes, and the policy degenerates to nearly-fixed play (85–129 switches per week against RA-TS-F’s ~ 850 ; regret $3.6\text{--}5.9\times$ worse, Table 4).

Fig. 4 shows the mechanism in the cleanest possible setting: an instrumented *toy* environment with synthetic rewards drawn from three recurring regimes that switch *discretely* (abrupt

boundaries, equal reward levels, a single fixed gap), using the same detector and matching calibration as the production runs. The CUSUM statistic of the played arm rests near zero within a regime, rises sharply at each hidden switch, and every alarm is followed by a brief ACQUIRE probe (shaded, a few steps each) after which the correct regime is recognised and its posterior resumed. Crucially, because the regimes here are clean and discrete, the library settles at exactly $J = 3$ —one entry per true regime—and the tracked regime $g(t)$ matches the truth $z(t)$ throughout.

This is deliberately *not* what happens in the physical environment, and the difference is instructive. In the detector simulations the library grows beyond the three nominal regimes ($J = 8$ over one week, $J = 17$ over six months; Figs. 6 and Sec. 7.6) for two reasons absent from the toy. First, the regime gaps are tiny and unequal (down to 0.005 in R_2) and identification runs on only $n_{\min} = 3$ noisy probe samples per arm, so a recurring regime whose noisy shape estimate misses its match margin occasionally finds a duplicate entry rather than resuming the original. Second, and more fundamentally, the physical regime transitions are *continuous ramps*, not the toy’s instantaneous switches (Section 2.2); the mid-ramp blended states are genuinely distinct reward shapes and the algorithm faithfully files them as short-lived extra entries (Sec. 7.6 quantifies this). Neither effect degrades performance—duplicate entries converge to the same best arm and transition entries cover genuinely ambiguous windows—which is why RA-TS-F attains the lowest regret despite carrying a “dirtier” library than the toy’s clean $J = 3$. The toy thus isolates the mechanism; the real runs show it remains effective when the discrete-regime idealisation is only approximate.

7.4 Experimental protocol

All policies are run on *identical*, pre-generated noise realisations (common random numbers): the regime schedule, injected suspension noise, input power, and per-window sensing noise are shared sample-by-sample, so policies differ only through their choices, and cumulative-reward differences are paired. The controller is held fixed within each $T_{\text{hold}} = 100\text{ s}$ window; switching uses the 2 s bumpless handover; the reward filters run continuously across the whole horizon (Section 5). We use three benchmarks: two one-week environments differing in storm frequency (*r1long*: one R_1 event per day on average; *r1freq*: two per day—roughly twice the regime switches), and one six-month environment ($1.55 \times 10^7\text{ s}$, $\sim 1.6 \times 10^5$ decisions, 181 storm events and ~ 3600 diurnal drift transitions). The figure of merit is cumulative regret against the Oracle; we also report the total number of controller switches and the fraction of windows in which a *near-optimal* controller is played. We call a controller near-optimal in a window when its expected per-window reward is within $\varepsilon = \frac{1}{2}\hat{\sigma} \approx 0.021$ of the best achievable (with $\hat{\sigma}$ the calibrated per-window reward noise). This is a more informative accuracy metric than the strict optimal-arm rate because two controllers are genuinely near-tied in some regimes—most sharply C_1 and C_2 in R_2 , gap 0.005 (Table 2)—so playing the runner-up there costs essentially no regret; ε sits an order of magnitude below every *costly* confusion (gaps ≥ 0.08), so the metric counts only genuine mistakes and, unlike the strict rate, tracks regret rather than contradicting it.

7.5 Results: one-week benchmarks

RA-TS-F attains by far the lowest regret on both caches (Table 4, Fig. 5): 40.7 on *r1long* and 25.8 on *r1freq*, against 125–217 for every baseline. Two features of this gap deserve emphasis. First—important for robustness—RA-TS-F’s advantage *grows* on the harder, more frequently switching cache, where every baseline stagnates or degrades. This is the recurrence dividend: a policy without regime memory re-pays the identification cost at every regime episode, so doubling the episode rate doubles that loss, while RA-TS-F’s per-episode cost is a few probe windows followed by a posterior resume. Second, the failure modes of the baselines are exactly the difficulties of Section 7.1: the rule-based operator policy degrades sharply with switching

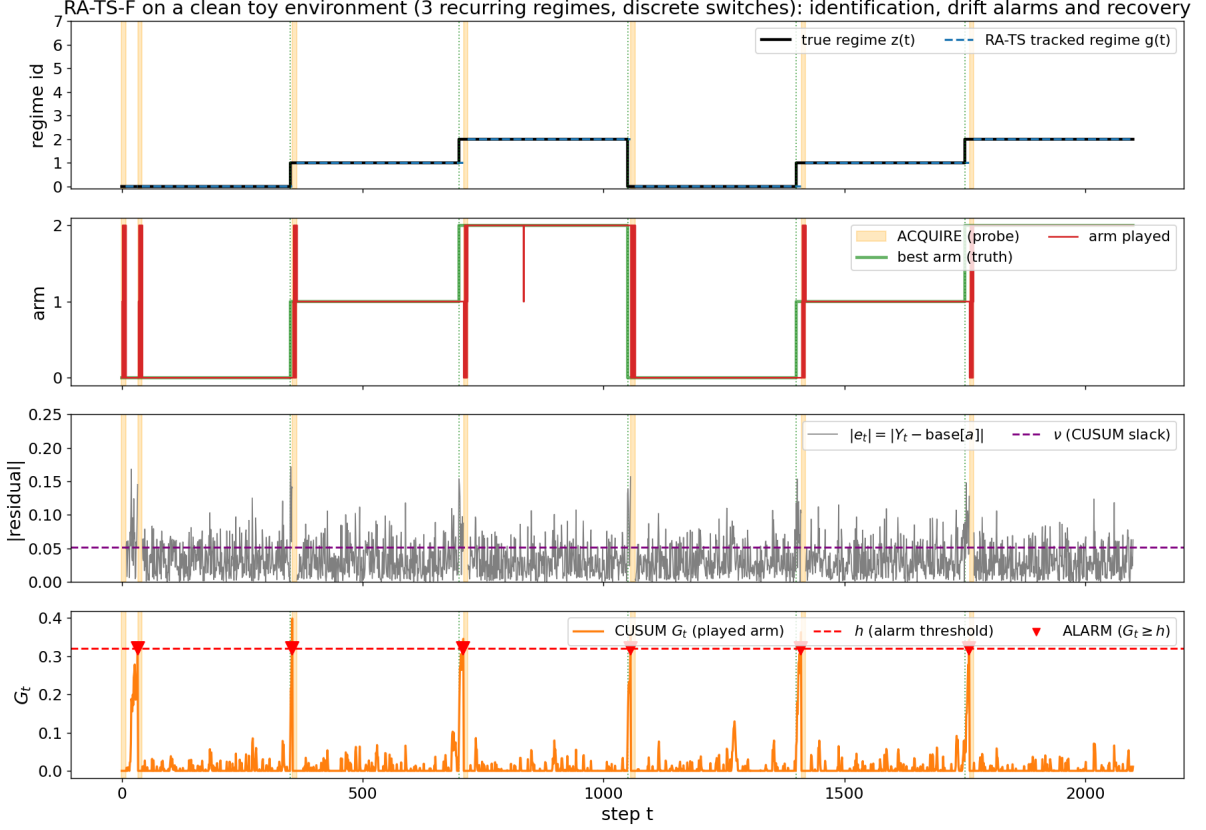


Figure 4: RA-TS-F mechanics on a *clean toy* environment: synthetic rewards, three recurring regimes with discrete switches and equal reward levels, same detector/matching calibration as the production runs. From top: true regime $z(t)$ and tracked regime $g(t)$ (they coincide, and the library settles at exactly $J = 3$); played arm vs. true best arm, with ACQUIRE probes shaded; per-step detector residual $|r - b_a|$ against the CUSUM slack ν ; CUSUM statistic G with threshold h and alarms. Every hidden switch drives G over threshold within a few decisions; the subsequent ACQUIRE probe lasts only a handful of windows before the returning regime is recognised (shape match) and its posterior resumed. On the physical environment the same mechanism runs but the library grows beyond $J = 3$ (tiny unequal gaps and continuous transitions; see text), without loss of regret.

frequency (125 $\rightarrow$ 176), the price of level-blind, schedule-driven re-checking (difficulty 1); and the passively forgetting bandits are barely better than the best fixed arm on either cache—generic forgetting cannot exploit recurrence (difficulty 3), and by continually discarding evidence it never resolves the small gaps (difficulty 2). Measured against the best fixed controller (C_0 , regret 155.9/151.2), RA-TS-F captures 74%/83% of the achievable Oracle advantage.

The two purity-preserving ablations of RA-TS-F probe whether library contamination—post-switch samples committed to the old regime during the detection delay—matters in practice. Rollback (FR) retroactively un-commits the change-point run on each alarm; delayed commit (FQ) quarantines samples for $q = 10$ rounds. Neither helps: both match RA-TS-F on r1long and lose on r1freq (where FQ’s stale posterior and FR’s discarded evidence cost most), so the plain tolerance-based design is not merely simpler but empirically the best of the three.

Fig. 6 dissects *how* RA-TS-F spends its budget on r1long. Probing (ACQUIRE) accounts for 9.4% of decisions and is concentrated in short bursts at the regime boundaries—between alarms the policy is pure warm-posterior exploitation. The library stabilises within two days and no new entries are created for the remaining five. Its final size is $J = 8$ rather than 3:

Table 4: One-week benchmarks: cumulative regret vs. the Oracle (lower is better) and number of controller switches, on the identical noise realisation per column. ~ 6000 decisions per run. Oracle cumulative reward: 2153.9 (rllong), 2138.6 (rlfreq).

policy	regret		switches	
	rllong	rlfreq	rllong	rlfreq
RA-TS-F	40.7	25.8	896	820
RA-TS-FR (rollback)	40.6	37.8	959	1188
RA-TS-FQ (delayed commit)	41.1	39.3	1058	1020
RA-TS (no coverage floor)	147.0	152.2	85	129
Rule-based	125.1	175.6	68	64
D-UCB	159.6	154.1	1985	2047
Thompson (discounted)	176.8	170.6	33	31
Fixed- C_0	155.9	151.2	0	0
Fixed- C_1	191.0	188.7	0	0
Fixed- C_2	217.2	216.1	0	0

Table 5: Six-month run (180 days, 155,519 decisions, identical noise realisation for all policies): cumulative regret vs. the Oracle, fraction of windows playing a near-optimal controller (within $\frac{1}{2}\hat{\sigma}$ of the best, Section 7.4), and number of switches. Oracle cumulative reward 57106.4.

policy	regret	frac. near-opt.	switches
Oracle	0	0.997	865
RA-TS-F	861.5	0.936	16,959
Rule-based	3702.9	0.626	1683
Fixed- C_0	3872.7	0.508	0
Thompson (disc.)	4095.3	0.509	437
Fixed- C_1	5235.0	0.522	0
D-UCB	5319.3	0.521	44,255
Fixed- C_2	6005.2	0.501	0

with $n_{\min} = 3$ probe samples per arm, the shape-estimate noise is comparable to the small calibrated separation Δ_{sh} , so a recurring regime occasionally fails its match margin and finds a duplicate entry. Duplicates are benign—each converges to the same best arm, and the regret cost of maturing one is a few windows—but they mark the price of identifying from only $n_{\min}K$ noisy samples; we return to this at the six-month scale.

7.6 Results: six-month run

The six-month run tests whether the one-week picture survives scale, slow parameter drift, and hundreds of regime episodes. It does (Table 5, Figs. 7–8): RA-TS-F’s regret of 861.5 is $4.3\times$ lower than the best alternative (the rule-based policy) and up to $7\times$ lower than the rest, capturing 78% of the Oracle’s advantage over the best fixed controller—quantitatively consistent with the one-week ratios, i.e. the advantage is not a short-horizon artefact. The mean per-window regret is $0.0055 \approx 0.13\hat{\sigma}$. The switch counts show that switching a lot is neither necessary nor sufficient: D-UCB switches 44k times for the worst learner regret, the rule-based policy only 1683 times, and RA-TS-F sits between (17k, 11% of decisions) with its exploration event-triggered by the drift detector rather than scheduled. The near-optimal fraction sharpens the picture: RA-TS-F plays a near-optimal controller in 93.6% of all windows, against $\leq 63\%$ for every baseline and

Regret accumulated over the 7-day run (lower = better; flatter = fewer mistakes)

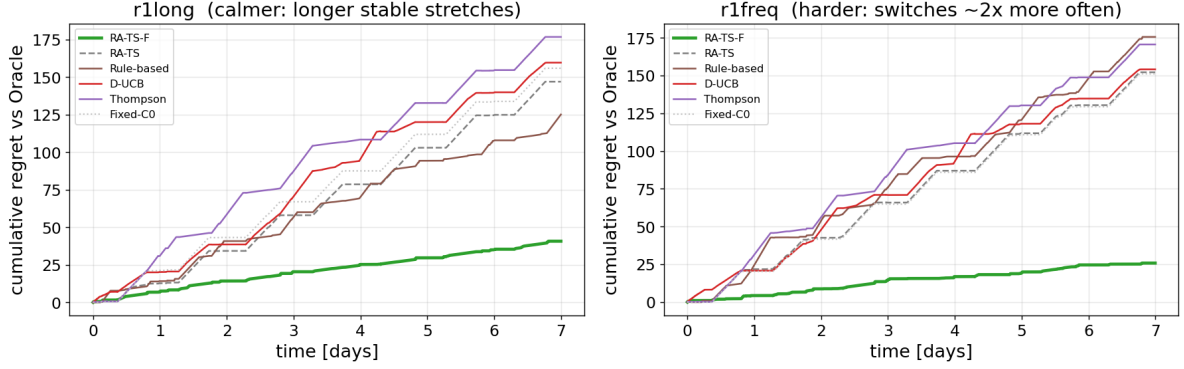


Figure 5: Cumulative regret against the Oracle over the two one-week benchmarks. RA-TS-F (green) accumulates regret at a rate 3–6 $\times$ lower than every baseline, and its advantage widens on the harder cache (right). RA-TS without the coverage floor (grey dashed) stalls and tracks the fixed-controller slope (dotted), its stepped growth marking mis-handled regime episodes; the discounted bandits (red, purple) barely improve on the best fixed controller; the rule-based policy (brown) is competitive on the calm cache but degrades under frequent switching.

only $\sim 51\%$ for any fixed controller. (The strict optimal-arm rate is misleading here—it scores RA-TS-F at 0.48, tying Fixed- C_0 , because R_0 dominates the duty cycle and it gives no credit for playing the near-tied runner-up in R_2 ; the near-optimal rate removes that artefact and cleanly separates the policies in the same order as regret.) Notably each fixed controller is near-optimal in almost exactly the fraction of time its regimes are active (~ 0.5)—strong on its own regimes, wrong elsewhere—whereas RA-TS-F is near-optimal almost everywhere, which is precisely the behaviour an autonomous selector must provide.

Two further observations from the six-month diagnostics. First, the regret slope of RA-TS-F is stable across the entire horizon (Fig. 7, right): the resumed posteriors do not go stale despite the environment’s slow within-regime drift (~ 3600 drift transitions), because the CUSUM alarm simply triggers re-acquisition whenever a stored baseline stops matching. Second, the library grows to $J = 17$ entries over 180 days. Diagnosing them individually shows the excess is *not* primarily duplicate noise: six entries are duplicates of the R_0/R_1 clusters (the same $n_{\min}$ -probe effect seen at week scale), but eight are short-dwell entries that map onto genuine *transition states*—windows in which the environment is mid-ramp between regimes (13.5% of tracked time; even the Oracle’s reward drops from 0.40 to 0.19 there). The algorithm, built for discrete recurring regimes, faithfully discovers that this environment’s switches are continuous ramps and files the ramp as extra regimes. This is a bookkeeping artefact with no measurable regret cost, but it delineates cleanly where the discrete-regime model of the environment ends.

8 Discussion

The infrastructure and measurements above delimit what an online controller-selection policy can and cannot achieve in this setting. The regime-controller problem is well posed, with the one-to-one mapping holding in all three regimes and a non-saturating reward; the per-regime separation is largest in the nominal regime and smallest (though still favouring the designed controller) in the extreme regime, so the achievable advantage of a selection policy is modest and accrues across all regimes. At the same time, with a realistic bumpless handover the cost of switching is small, so regret is governed primarily by *which* controller is chosen rather than by the act of switching. These two observations frame the bandit comparison of Section 7: the relevant question is how quickly and reliably a policy identifies the regime-optimal controller as

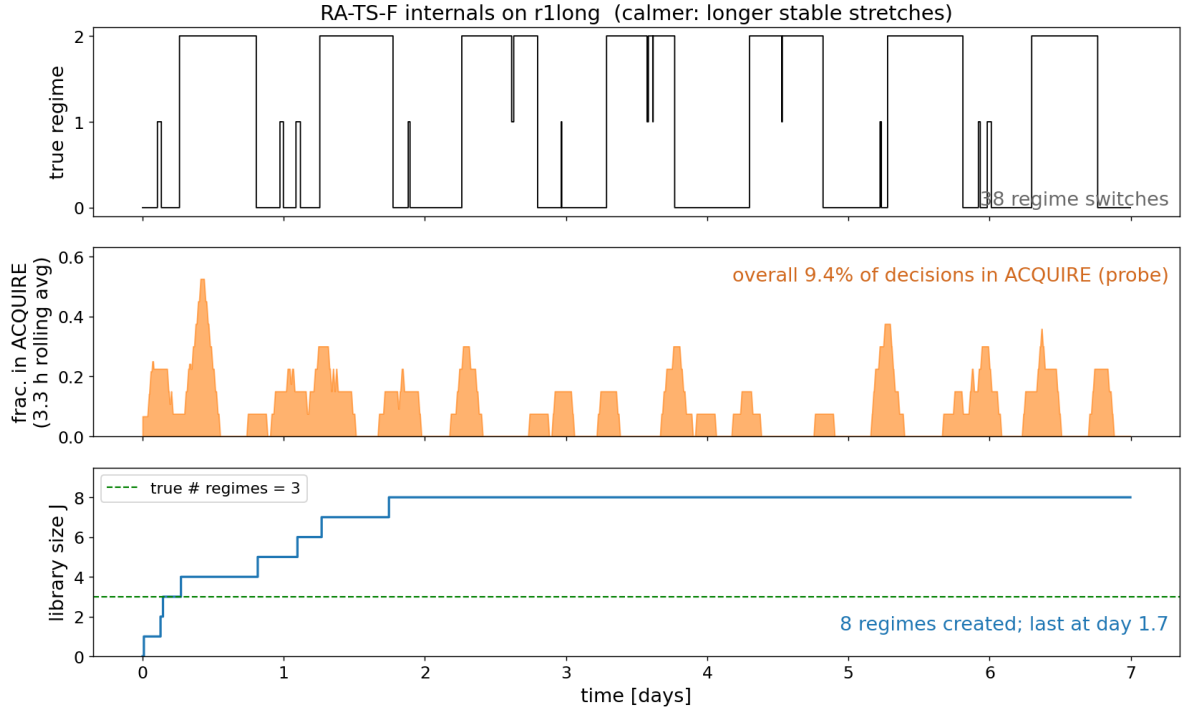


Figure 6: RA-TS-F internals over the r1long week (three stacked panels). From top: (i) the true regime sequence (38 switches); (ii) the fraction of decisions spent in ACQUIRE (the re-identification probe), shown as a rolling average over a 3.3 h (120-decision) sliding window—probing is 9.4% of decisions overall and occurs in short bursts localised at the regime boundaries, so between boundaries the policy is pure warm-posterior exploitation; (iii) the library size J , which stabilises by day 1.7 at $J = 8$ (the three true regimes plus noise-split duplicates, see text) and adds no further entries over the remaining five days.

the environment drifts, with switching essentially free.

The bandit results answer that question with a clear structural message: exploiting *recurrence* is what matters. Resuming per-regime knowledge beats re-learning it at every episode, and beats forgetting it (the discounted baselines) by a wide margin. Just as telling is what proved unnecessary: our earlier design iterations carried scheduled probes, explicit identification phases, and statistical certification tests, each added to fix a real failure mode; all of them became removable once identification was made emergent from Thompson sampling itself, guarded only by a bounded coverage floor—and every removal improved regret. The floor is the one non-negotiable ingredient: without it, Thompson sampling’s own efficiency starves the identification signal and the policy stalls ($3.6\text{--}5.9\times$ worse regret, Table 4). Its cost is bounded ($n_{\min}K$ windows per re-acquisition, 9.4% of decisions in practice). A practical corollary for detector operations: the reward signal that suffices for all of this is computable online from existing error and control channels; no auxiliary environmental sensing is required, although incorporating such context (e.g. seismometer bands or earthquake early warning) as side information is a natural extension.

For future learned (e.g. neural-network) controllers two elements must be revisited: the reward separation between arms is expected to grow (learned controllers specialise more aggressively than a gain family), which makes the selection problem easier; but the near-free switching result of Section 6 relies on the gain-family structure and must be re-measured, since a convex blend of two stabilising controllers of different shape is not guaranteed stabilising mid-ramp. The RA-TS-F layer itself is agnostic to both changes.

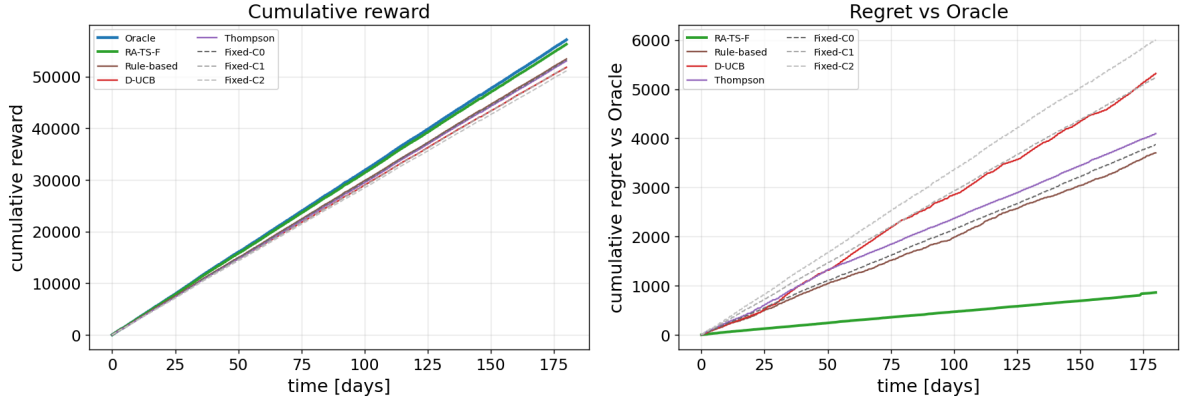


Figure 7: Six-month run: cumulative reward (left) and cumulative regret against the Oracle (right) for all policies. RA-TS-F’s regret grows at a near-constant, far shallower slope than every alternative throughout the 180 days—there is no late-horizon degradation, saturation of the advantage, or drift-induced breakdown.

9 Conclusion

We posed the operational practice of switching angular-control filters as an online controller-selection problem and built the infrastructure to study it rigorously: a validated time-domain plant, three physically motivated operating regimes with a designed one-to-one regime–controller mapping, a calibrated multi-band reward computed online from the loop’s own signals, and a common-random-numbers benchmark spanning one-week and six-month horizons. Three findings stand out. First, the selection problem is well posed but genuinely hard: the reward distributions of the candidate controllers overlap strongly, two of the three regimes are separated by gaps below the per-window noise, and the regime identity is carried by the cross-arm reward *shape* rather than its level. Second, the apparent cost of switching controllers is dominated by the handover implementation; with a bumpless, parallel-bank handover—standard practice in detector digital control—switching is nearly free, removing the main physical objection to autonomous switching. Third, a deliberately minimal policy, Recurrence-Aware Thompson Sampling with a bounded coverage floor (RA-TS-F), outperforms every alternative tested—discounted bandits, an operator-like rule, and all fixed controllers—achieving 74–83% of the oracle’s achievable advantage over the best fixed controller, with a regret slope that remains constant over 180 simulated days. The combination of a cheap handover, an online reward from existing channels, and a simple recurrence-aware policy makes autonomous alignment-configuration selection a realistic operational prospect; extending the same framework to learned controllers on hardware-validated setups is the natural next step.

Acknowledgements

[TODO: funding and computing acknowledgements.]

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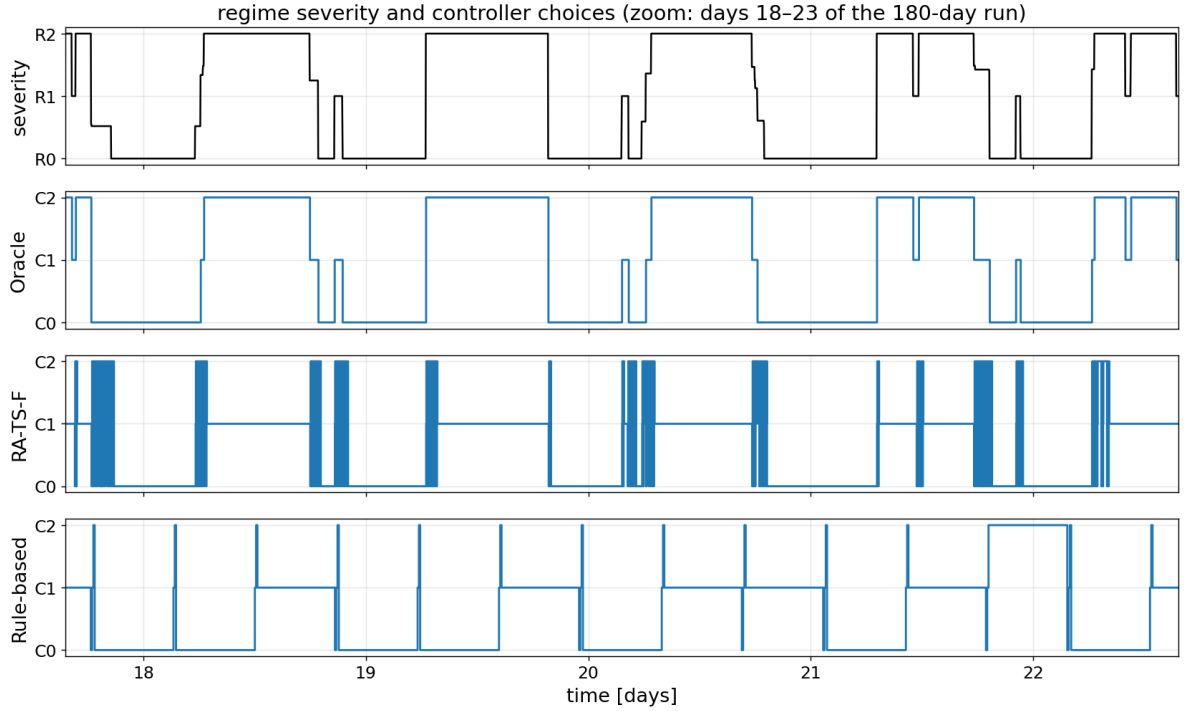


Figure 8: Six-month run, zoomed to a representative 5-day window (the full 180-day timeline is an unreadable dense band). Top: regime severity ($R_0/R_1/R_2$). Below: controller choices of the Oracle, RA-TS-F, and the rule-based policy. RA-TS-F tracks the regime structure closely—it plays C_0 through the R_0 stretches and captures the R_1 spikes, with brief vertical bursts marking the acquisition probes at each boundary. Through the R_2 stretches it settles on C_1 rather than the Oracle’s C_2 : the two are near-tied there (gap 0.005, Fig. 1), so Thompson sampling does not distinguish them and the choice costs essentially no regret—this is precisely why the near-optimal fraction, not the strict optimal-arm rate, is the meaningful accuracy metric (Table 5). The rule-based policy switches rarely (1683 total) and visibly lags the severity pattern, its re-checks being schedule- rather than event-driven.

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